Electrochimica Acta 195 (2016) 85–93

Contents lists available at ScienceDirect

Electrochimica Acta

journal homepage: www.elsevier.com/locate/electacta

Accounting for the concentration dependence of electrolyte diffusion coefficient in the Sand and the Peers equations

S.A. Mareev[a], D.Yu. Butylskii[a], A.V. Kovalenko[a], A.V. Petukhova[a], N.D. Pismenskaya[a], L. Dammak[b], C. Larchet[b], V.V. Nikonenko[a,*]

[a] Kuban State University, 149 Stavropolskaya st., 350040 Krasnodar, Russia
[b] Institut de Chimie et des Matériaux Paris-Est, UMR 7182CNRS—Université Paris-Est, 2 Rue Henri Dunant, 94320 Thiais, France

ARTICLE INFO

Article history:
Received 15 December 2015
Received in revised form 15 February 2016
Accepted 16 February 2016
Available online 20 February 2016

Keywords:
membrane
voltammetry
chronopotentiometry
limiting current
transition time
numerical simulation

ABSTRACT

The equations for calculating the diffusion limiting current, $i_{\lim}$, in steady-state voltammetry (known as the Peers equation in membrane science) and the transition time, τ, in chronopotentiometry (the Sand equation) are broadly used in electrode and membrane electrochemistry. The applicability of these equations is limited because they are deduced under the assumption of a constant diffusion coefficient. However, within the diffusion boundary layer, the diffusion coefficient, D, varies between the values corresponding to the bulk solution (D_b), and the infinitely dilute solution (D_0) near the electrode or membrane surface. In this paper, we explore two models, which account for the concentration dependence $D(c)$ in order to generalise the above fundamental equations. We show that the correct value of $i_{\lim}$ can be found via solution of a 2D model, while to find τ, a 1D non-stationary model is sufficient. Generally, the dependence of $i_{\lim}$ on the bulk concentration deviates from the proportionality. The similar situation occurs with the proportionality of τ to the squared concentration in the Sand equation. We show that the numerical solutions for $i_{\lim}$ and τ can be presented in the forms analogous to the Peers and the Sand equations, respectively, but with an additional correction factor. In particular, an effective diffusion coefficient, D_{ef} (an average between D_b and D_0) has to be introduced in the case of Sand equation. Comparison of our theoretical prediction with experimental chronopotentiograms confirms the necessity of taking into account the $D(c)$ dependence.

1. Introduction

The equation expressing the limiting current density, $i_{\lim}$, across an electrode or ion-exchange membrane in the case of film diffusion kinetics may be present as follows [1]:

$$i_{\lim} = \frac{FDc_0}{\delta(T_1 - t_1)} \quad (1)$$

where D is the electrolyte diffusion coefficient related to the individual ion diffusion coefficients D_k: $D = 2D_1D_2/(D_1 + D_2)$ (in the case of 1:1 electrolyte), c_0 is the electrolyte concentration (equiv dm^{-3}) in a stirred bulk solution, δ is the diffusion boundary layer (DBL) thickness, F is the Faraday number, t_1 is the counterion transport number in solution, T_1 is the so called integral [2,3] or effective [4,5] counterion transport number across the solid/solution interface. In the case of electrode, $T_1 = 1$. In membrane science, according to the definition given in [2–5],T_k is the fraction of (faradaic) electric current transported through the membrane/solution interface by ion k:

$$T_k = \frac{z_k F (j_k)_s}{i}, \quad (2)$$

where i is the current density, z_k and $(j_k)_s$ are the charge number of ion k and its flux through the interface, respectively. We consider a binary electrolyte, subscript k = 1 or 2 are for the counterion or the co-ion, respectively.

Commercial ion-exchange membranes are high permselective to counterions so that T_1 is usually only few less than 1 [6,7], that is most of the current is carried by counterions, while the counterion transport number t_1 in the solution is close to 0.5.

Eq. (1) for the case of ideally permselective membrane was first deduced by Peers [1]. The form involving the transport number in the membrane was proposed by Rosenberg and Tirrell [8].

Generally, the electrolyte diffusion coefficient, D, involved in the Nernst-Planck equation, which is on the basis of Eq. (1), is a function of concentration. Since at the limiting current density, the

* Corresponding author. Tel.: +7 861 2199 573; fax: +7 861 2199 517.
E-mail address: mareev-semyon@bk.ru (S.A. Mareev).

http://dx.doi.org/10.1016/j.electacta.2016.02.098
0013-4686/

concentration in the diffusion layer varies from a value related to the bulk solution, c_0, to a value close to zero at the electrode or membrane surface, D varies between the values corresponding to the bulk (D_b), and the infinitely dilute solution (D_0). In particular, a question appears, if Eq. (1) remains valid in the case of variable D? Moreover, even if this equation stays valid, what value of D should be used in practical applications? These questions apparently arose many times when Eq. (1) was applied for reaching different goals: to find δ from known i_{lim} [8], to evaluate i_{lim} [9], to determine T_1 [10]. There are studies where D value was assigned to infinitely dilute solution, D_0 [9,11], in others [12–14], it was assigned to the bulk solution concentration, D_b.

A similar problem occurs when applying the Sand equation [15], written here in the form first adapted to membrane transport by Lerche and Wolf [12] and later by Krol et al. [13]:

$$\tau = \frac{\pi D}{4}\left(\frac{c_0 z_1 F}{T_1 - t_1}\right)^2 \frac{1}{i^2}, \qquad (3)$$

where τ is the so called transition time. Note that in original papers [12,13] instead of T_1 it was the migration transport number in the membrane, $\bar{t}_1$ (which is the fraction of electric current transported by the counterion in a milieu under the condition of zero concentration gradient). However, certainly, the authors [12,13] had in mind that the total counterion transfer in the membrane should be involved (including migration, diffusion, convection), while the contribution of migration is dominant.

In chronopotentiometric (ChP) measurements under a given direct current density $i > i_{lim}$, the transition time determines the time needed for the electrolyte concentration at membrane (or electrode) surface to attain such a low value where the change in the mode of ion transfer occurs. In electrode systems, this low concentration conditions a threshold overvoltage corresponding to the onset of an additional electrochemical reaction. In membrane systems, new charge carriers, H^+ and OH^- ions can be generated in water splitting reaction at the depleted membrane interface [16–18]. Otherwise, current-induced convection (gravitational

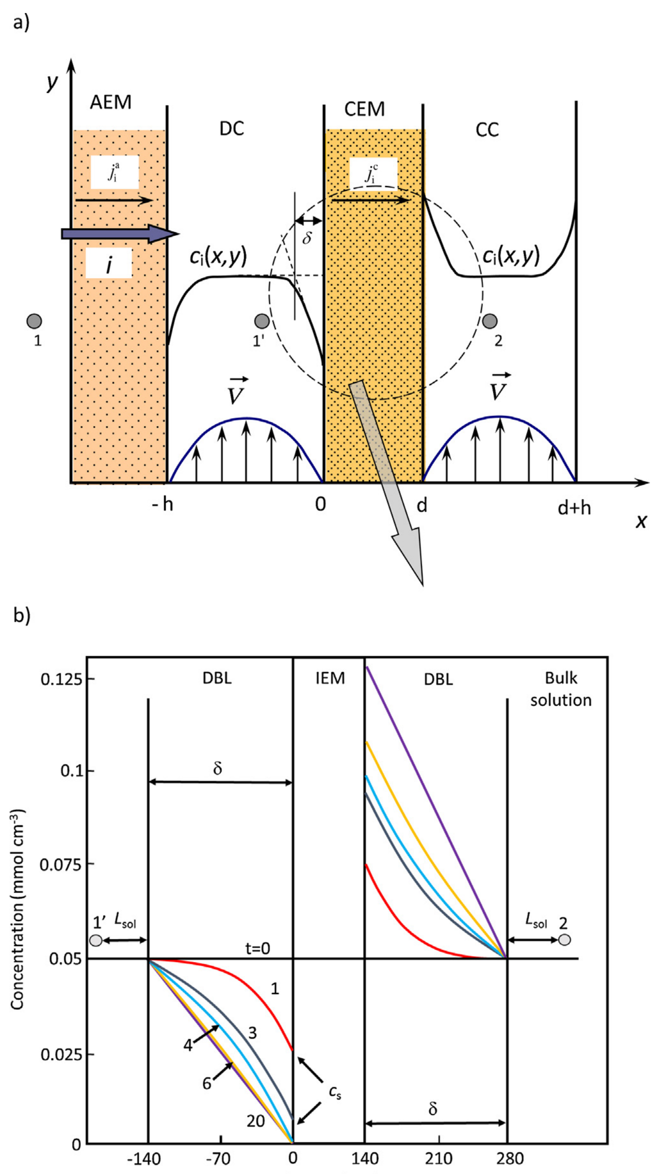

Fig. 1. a: Scheme of an electrodialysis cell with an anion-exchange (AEM) and a cation-exchange (CEM) membranes; DC and CC are desalting and concentrating compartments, respectively. The points 1, 1′ and 2 show where the tips of Luggin's capillaries may be placed. b: Concentration profiles development in diffusion boundary layers (DBL) adjacent to a CEM; the time in seconds is shown near the curves. The profiles are calculated using Eqs. (35)–(37), (41) for c_0 = 0.05 M NaCl and current density i = 21.25 mA cm^{-2}.

convection [19–22] and/or electroconvection [23–26]) can enhance ion transfer. After passing through the transition state, the system reaches a (quasi) steady state, where the potential difference does not change (or changes only slightly) over time. The new mechanisms gives rise to overlimiting conductance: on the I–V curve, the current values related to the steady state are above the line $i = i_{\lim}$ determined by the Peers Eq. (1).

The application of the Sand equation involves the same difficulty as in the case of the Peers equation: does this equation remain valid, when D is variable, and what value of D should be used in calculations: D_0 or D_b or an effective coefficient D_{ef}? While Eq. (3) is obtained under the assumption $D = const$, it is largely applied in the cases where D varies with distance between D_0 and D_b. The Sand equation was used for determination of the electrolyte diffusion coefficients in the bulk solution in electrode [27,28] and in membrane [29] systems by using the measured value of τ and the known (or estimated) ion transport numbers in the solution and the membrane.

It should be noted that the problem of determination of the D value in electrode systems can be solved in many cases by using the infinitely dilute solution ($D = D_0$), or introducing a base electrolyte [30]. In the last case, D is determined by the total ionic strength conditioned by the base electrolyte. However, the second approach is not applicable to membrane systems because the current flow affects the concentration of all ions in solution [2,4,5]. One of the important problems of the membrane electrochemistry is the understanding of membrane behavior as depending on the external solution concentration, c_0. It is clear that this problem cannot be solved using only the diluted solution. In particular, when determining the ion transport number in the membrane using the experimental transition time [31], knowing of the exact value of D is necessary: according to Eq. (3), when D is increased by 10%, the co-ion transport number (T_2) decreases by 0.025, when τ is fixed. This error may be inadmissible if T_2 takes a value of the same order. As an example, let us mention that the typical value of T_2 in a heterogeneous cation-exchange membrane (MK-40 equilibrated with 1 M NaCl) is 0.03, while $T_1 = 0.97$ [7].

As far as we know, there are only two publications made by Kubíček [32,33] where the concentration dependence of diffusion coefficient was taking into account in theory of transition processes. In these papers [32,33], a linear function expressing $D(c)$ was applied, and approximate equations generalizing the Cottrell equation in chronoamperometry [32] and the Sand equation in chronopotentiometry [33] were obtained.

In this paper, we explore two models, which account for the concentration dependence $D(c)$ in order to generalise the above fundamental equations. We show that the correct value of $i_{\lim}$ can be found via solution of a 2D model, while to find τ, a 1D non-stationary model is sufficient. It will be verified, if the dependence of $i_{\lim}$ on the bulk concentration deviates from the proportionality. The similar situation occurs with the proportionality of τ to the squared concentration in the Sand equation. We show that the numerical solutions for $i_{\lim}$ and τ can be presented in the forms analogous to the Peers and the Sand equations, respectively, but with an additional correction factor. We use a polynomial approximation $D(c)$, which is valid in a large concentration range, unlike the linear approximation used in [32,33]. We find that in the Peers equation, D_0 should be used, however, the concentration dependence of the diffusion layer thickness should be accounted. In the case of the Sand equation, it is possible to find such an effective value, D_{ef}, an average between D_b and D_0, the substitution of which in the Sand equation gives the same value of transition time as the numerical calculation. Comparison of calculations using $D = const$ and variable $D(c)$ with experimental data is discussed.

2. Concentration polarization. Limiting current density

Let us consider an elementary fragment of an electrodialysis apparatus (Fig. 1), a so called cell pair, which consists of a cation- and an anion-exchange membranes forming a desalting and a concentrating compartments. The same binary electrolyte solution enters both compartments.

An electric current is directed perpendicular to the membrane surface. Immediately after switching on the current (time $t = 0$, Fig. 1b), the migration flux of counterions from the interface towards the membrane bulk (where their transport number is close to 1) significantly exceeds their flux from the solution bulk (the transport number is near 0.5) towards the interface. As a result, counterion (and co-ion) concentration in the adjacent to the membrane solution layer decreases at one membrane side and increases at the other (the concentration polarization phenomenon). This leads to appearance of ion diffusion from the solution bulk towards the membrane, which tends to equalize the incoming and outgoing fluxes at the interface. Thus generally, the ion flux in the near-interface solution involves migration, diffusion and convection (the forced or current-induced ones); it is described by the "extended" Nernst-Planck equation:

$$\vec{j}_k = -D_k\left(gradc_k + z_k c_k \frac{F}{RT} grad\varphi\right) + c_k\vec{V}, \tag{4}$$

where D_k and c_k are the diffusion coefficient and the concentration (in mole dm^{-3}) of ion species k, respectively, $\vec{V}$ is the fluid velocity (cm s^{-1}), φ is the electric potential, R is the gas constant, T is the absolute temperature.

Eq. (4) is added by the equation linking the current density with ionic fluxes, the electroneutrality condition and the continuity equation:

$$\vec{i} = F\sum z_k\vec{j}_k, \tag{5}$$

$$\sum z_k c_k = 0, \tag{6}$$

$$\frac{\partial c_k}{\partial t} = -\mathrm{div}\vec{j}_k. \tag{7}$$

Only the faradaic current is considered in this paper. The displacement current should be taken into account in the description of phenomena where the electric double layers at the interfaces plays an important role, such as high frequency impedance [34–36]. In 0.002 M NaCl solution the time needed for establishing the steady-state concentration profiles in an equilibrium electric double layer estimated as L_D^2/D is less than 10^{-8} s, where L_D is the Debye length. The time scale related to the formation of an extended space charge, added to the usual one of a quasiequilibrium EDL in limiting current regime is estimated as $t_{SC} = (L_D^2/D)/(L_D/\delta)^{2/3}$ [34], where δ is the diffusion layer thickness. For the above conditions, t_{SC} is of the order of 10^{-5} s, that is five orders of magnitude less than the transition times.

2.1. Deduction of the Peers equation

Usually in hydrodynamics the no-slip condition is set on solid surfaces ($\vec{V}(x = 0, y) = 0$) [37]. This results in small contribution of the convective term in Eq. (4) near the membrane (or electrode) surface. Moreover, when considering the normal to the flat surface flux, in the absence of current-induced convection this term is negligible.

In case of binary electrolyte, the electric field intensity $\vec{E} = -grad\varphi$ expressed from Eqs. (4)–(6), has the form:

$$\vec{E} = \frac{RT}{F}\left[\frac{D_1 - D_2}{D_1 z_1 - D_2 z_2}\frac{gradc}{c} + \frac{\vec{i}}{(D_1 z_1 - D_2 z_2)cF}\right], \tag{8}$$

where $c = z_1 c_1 = -z_2 c_2$ is the electrolyte concentration (in eq dm^{-3}).

After substituting Eq. (8) in Eq. (4), we obtain the following transport equation:

$$\vec{j}_k = -D\text{grad}c_k + \frac{\vec{i}t_k}{z_k F} + c_k\vec{V}, \tag{9}$$

where the electrolyte diffusion coefficient, D, and the ion transport number, t_k, in solution are expressed as follows:

$$D = (z_1 - z_2)D_1 D_2/(D_1 z_1 - D_2 z_2), \tag{10}$$

$$t_k = z_k D_k/(z_1 D_1 - z_2 D_2). \tag{11}$$

At the membrane/solution interface, in the conditions of the absence of charging processes, there is continuity of the normal flux component even in a non-stationary process [22,38,39]:

$$(j_k)_s = \left(-D\frac{\partial c_k}{\partial x} + \frac{it_k}{z_k F}\right)_{x=0-} = \frac{iT_k}{z_k F}, \tag{12}$$

where $x = 0$- denotes the left-hand limit to the interface, that is from the solution side (Fig. 1b); the last part of Eq. (12) refers to the interface from the membrane side and responds to the definition of the effective transport number expressed by Eq. (2). Note that the solution is assumed electrically neutral; hence, the domain under consideration is outside of the interfacial electric double layer.

It follows from Eq. (12) that:

$$\left(\frac{\partial c_1}{\partial x}\right)_{x=0-} = \frac{c_{10} - c_{1s}}{\delta} = \frac{i(T_1 - t_1)}{z_1 FD}, \tag{13}$$

where c_{10} and c_{1s} are the counterion concentrations in the bulk solution and at the membrane surface, respectively. According to Eq. (13), δ is the distance between the membrane surface and the intersection point of the tangents drawn to the concentration profile at the interface and in the bulk solution (Fig. 1a).

As we can see from Eq. (13), $(\partial c_1/\partial x)_{x=0-}$ depends only on the current density and does not explicitly depend of time. However, there is an implicit dependence. In chronopotentiometry when a constant current density i is applied, c_{1s} decreases with time (Fig. 1b), hence, δ increases. c_{1s} and δ stop to change when the system reaches a steady state. With increasing i, the steady state concentration of counterion (as well as that of co-ion) at the depleted interface decreases. When c_{1s} is close to zero (more precisely it becomes much smaller than c_{10}), the current density reaches its limiting value (i_{lim}). Assuming that $c_{1s} << c_{10}$ in Eq. (13), we obtain the Peers equation (1) with $c = z_1 c_1$.

The deduction above shows that the application of boundary condition (13) and the replacement of the concentration gradient at the interface by the $c_{10} - c_{1s})/\delta$ ratio are the key operations when obtaining Eq. (1). As the derivative is taken at the membrane surface in the depleted solution at $i = i_{lim}$, $c_{1s} << c_{10}$, D must be referred to the infinitely diluted solution: $D = D_0$. However, δ is a function of the diffusion coefficient [37], which, in turn, depends on the concentration. Thus, the D/δ factor in Eq. (1) depends on the concentration not via the diffusion coefficient, but via δ. Then the problem of accounting for *D(c)* in the Peers equation becomes as follows: how to find the effective value of δ as a function of bulk concentration in a way that its substitution in the Peers equation gives the "correct" value of i_{lim}? Under the "correct" value we understand the value of i_{lim}, which gives an appropriate model taking into account the concentration dependence of D.

It is impossible to solve this problem using 1D simulation as in the models of this kind δ is a parameter, which should be found outside of these models. However, it is possible to find the answer by using 2D modelling. Below we apply a 2D diffusion-convection model for the membrane system shown in Fig. 1a.

3. 2D diffusion-convection model for generalizing the Peers equation

The simplest way to find an answer to the question above is the use of a 2D problem, which is valid only for the limiting current density. The model involves the Nernst-Planck vector equation, Eq. (9), with convective term. For the fluid flow between two parallel plates (without spacer) considered as laminar and steady, the normal velocity component is zero, $V_x = 0$, and the tangential component is described by the Poiseuille equation [37]:

$$V_y(x) = 6V_0\left(\frac{x}{h} - \frac{x^2}{h^2}\right), \tag{14}$$

where y is the longitudinal coordinate, V_0 is the average velocity, h is the distance between the plates.

In ED cells, the longitudinal components of the concentration gradient and the current density are assumed negligible in comparison with the normal ones. This assumption is common for underlimiting current densities [40,41], however, it cannot be accepted if current-induced convection is taken into account at overlimiting currents [25,42,43]. As we consider the case of limiting current in classical approach [40,41], the components of the fluxes are taken as follows:

$$j_{ix} = -D\frac{\partial c_i}{\partial x} + \frac{it_i}{z_i F}, \tag{15}$$

$$j_{iy} = c_i V_y(x). \tag{16}$$

Accounting for the concentration dependence of the diffusion coefficient is carried out using the Nernst- Hartley equation [44]:

$$D = D_0 g, \tag{17}$$

where D_0 is the diffusion coefficient at infinite dilution, g is the activity factor defined as [44]:

$$g = 1 + d(\ln y_{\pm})/d(\ln c), \tag{18}$$

where $y_{\pm}$ is the mean molar activity coefficient.

Application of Eq. (7) to Eqs. (15) and (16) for steady state $(\partial c_i/\partial t = 0)$ gives the equation similar to that studied in [37,40,41,45] at D = const:

$$D_0\frac{\partial}{\partial x}\left[g(c)\frac{\partial c}{\partial x}\right] = 6V_0\left[\frac{x}{h} - \left(\frac{x}{h}\right)^2\right]\frac{\partial c}{\partial y}. \tag{19}$$

The concentration dependence of transport numbers in solution, t_i entering Eq. (9), is not taken into account, since t_i is a function of the ratio of the diffusion coefficients, Eq. (11). As D_i for cation and anion change with concentration in a similar way, t_i depends on the concentration much less than D.

Here for the sake of simplicity, we remove index "y" for the tangential velocity, and use "c" for the electrolyte concentration, $c = z_+c_+ = -z_-c_-$.

The application of limiting current assumes a simple boundary condition at the membrane surface in the desalting compartment:

$$c(0, y)_{y>0} = 0 \tag{20}$$

$y = 0$ relates to the compartment entrance.

The input concentration at any x is equal to c_{0in}, which gives the initial condition:

$$c(x, 0) = c_{0in}. \tag{21}$$

Eqs. (12) and (13) expressing the continuity of fluxes have a general nature and must held in the 2D model. The concentration gradients at the interface found numerically when solving boundary-value problem (19)–(21) allow us to compute the current density through the membrane as function of the longitudinal coordinate y, Eq. (13). As well, Eq. (13) allows calculation of the Nernst diffusion layer thickness, δ, as a function of y.

3.1. Case of variable D

Problem (19)–(21) has an approximate analytical solution at D = const first obtained by Lévêque [46] for heat transfer. This solution is presented in Supplementary Materials.

The numerical solution of problem (19)–(21) allows calculation of the Nernst diffusion layer thickness (through the tangents to the concentration profile) as a function of the channel length. It is convenient to introduce the dimensionless channel length

$$Y = yD/V_0h^2. \tag{22}$$

In the case of D = const, at $Y \le 0.01$ $(L \le 0.01V_0h^2/D)$, the approximation of the numerically calculated values δ_{loc} in coordinates $\delta_{loc} - Y^{1/3}$ (Fig. 2) gives

$$\delta_{loc} \approx 1.06hY^{1/3}. \tag{23}$$

Note that if summand 0.2 is omitted in Eq. (1S) (see Supplementary Materials) and the obtained equation is used together with Eq. (1), we find the following expression known in literature [41]

$$\delta_{loc} \approx 1.02hY^{1/3}. \tag{24}$$

Eq. (24) gives a lower value of δ_{loc}, which corresponds to the single-term overstating approximation of i_{limloc} (Supplementary Material).

Substituting Eq. (23) in the Peers equation, Eq. (1), gives the local current density as a function of the longitudinal coordinate y in the form:

$$i_{limloc} = 0.940\frac{FDc_{0in}}{h(T_1 - t_1)}\left(\frac{h^2\bar{v}}{yD}\right)^{1/3}, \tag{25}$$

The lower coefficient in single-term Eq. (25) in comparison with the binomial Eq. (1S) (Supplementary Material) is assigned to take into account the role of summand 0.2.

Integration of Eq. (25) over the channel length yields the average limiting current density:

$$i_{lim} = 1.41\frac{FDc_{0in}}{h(T_1 - t_1)}\left(\frac{h^2\bar{v}}{LD}\right)^{1/3}. \tag{26}$$

Equating the right sides of Eqs. (26) and (1), we find an expression for the average (or effective) over length L diffusion layer thickness:

$$\delta_{av} = 0.71hY^{1/3} = 0.71h\left(\frac{LD}{h^2\bar{v}}\right)^{1/3}. \tag{27}$$

There is a very good agreement between the values of i_{limloc} and i_{lim} calculated by using Eqs. (25) and (26) obtained from numerical simulation, and by the binomial Lévêque Eqs. (1S) and (2S), respectively. The highest relative divergence between the results given by Eqs. (1S) and (2S) occurs at very small Y, where the second term in Eq. (1S) is negligible. However, these values of Y relate to too short channels not of practical use.

When integrating the local current density, Eq. (25), we obtain Eq. (26) for i_{lim} in the same form. However, there is a multiplier between i_{lim} and i_{limloc} equal to 3/2. Hence, the value of i_{lim} in the channel of length L is 3/2 times higher than the value of i_{limloc} at $y = L$. Similarly, the value of average value of DBL thickness in the channel of length L is 3/2 times less of the value of δ_{loc} at $y = L$.

3.2. Case of variable D

In this paper we consider solutions of LiCl, NaCl and KCl of different concentrations. Concentration dependence of the activity factor for these electrolytes is shown in Supplementary Materials, Fig. 1S.

The approximation of experimental data [47], shown in Fig. 3 allows us to represent $g(c)$ as a polynomial function. For KCl, NaCl and LiCl, this function has a form, respectively [48]:

$$g(c) = 0.1097c^2 - 0.4277c^{3/2} + 0.7346c - 0.4766c^{1/2} + 0.9972, \tag{28}$$

$$g(c) = 0.1213c^2 - 0.4331c^{3/2} + 0.7666c - 0.4494c^{1/2} + 0.9946, \tag{29}$$

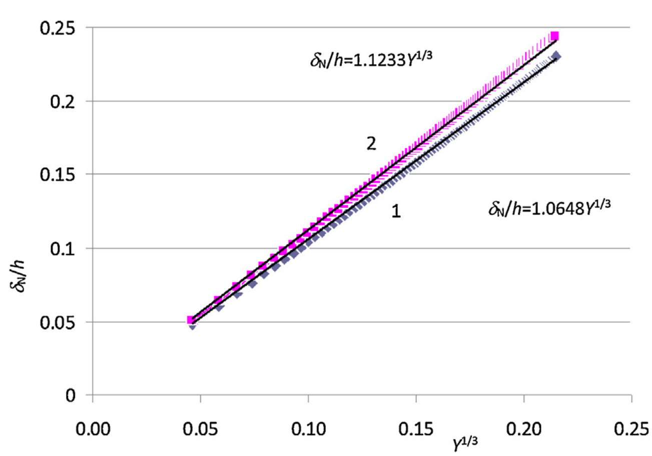

Fig. 2. Reduced Nernst DBL thickness (δ_{loc}/h) as a function of $Y^{1/3}$. The points are the results of the numerical calculation when assuming D = const (1), and when taken into account concentration dependence $D(c)$ according to Eqs. (17)–(19), in the case of a 0.2 M solution of KCl (2). The equations of approximating straight lines are shown.

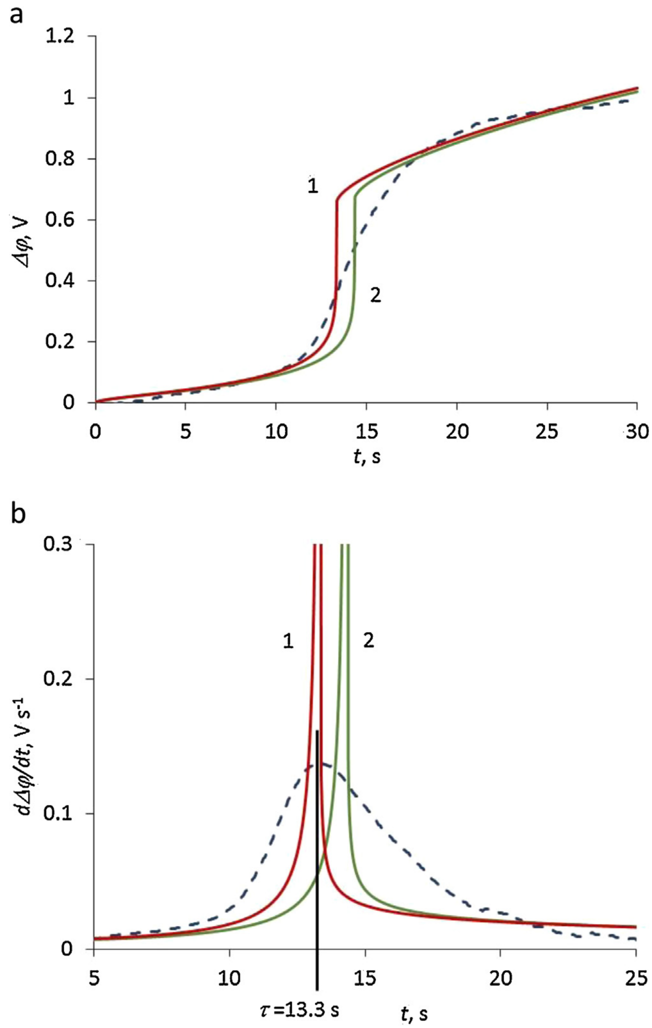

Fig. 3. Chronopotentiograms (a) and their time derivatives (b) for an AMX anion-exchange membrane in a solution of 0.2 M NaCl at a current density i = 47.7 mA cm^{-2}. Dashed lines are experimental data obtained using procedure described in [57], solid curves are calculated: when taking into account the dependence $D(c)$ (curve 1) and when assuming $D = D_0$ (curve 2).

$$g(c) = 0.2036c^2 - 0.6288c^{3/2} + 1.0658c - 0.474c^{1/2} + 0.9951, \tag{30}$$

the concentration of electrolyte, c, is in mol dm^{-3}; g, as it is seen from Eq. (18), is dimensionless.

3.3. Results of numerical computations

In the 2D simulation, we consider the case of KCl, for which the variation of the diffusion coefficient in the range of moderate concentrations (up to 0.5 mol dm^{-3}) is the most important among the other chlorides shown in Fig. 1S. Let the KCl concentration in the bulk solutions on the left and the right sides of the membrane is equal to 0.2 mol dm^{-3} at which the diffusion coefficient of KCl reaches its minimum. The transport number of counterions (potassium ions) in solution is equal to 0.4899, and assume to be 1 in the membrane, which is considered as an ideal selective cation-exchange membrane.

We use the approach described above: the value of δ_{loc} is calculated as a function of Y numerically when solving problem (17)–(21). The presentation of obtained data in the δ_{loc} vs. $Y^{1/3}$ coordinates allows obtaining the dependence in the $\delta_{loc} = hkY^{1/3}$ form, where coefficient k takes different values depending on the input concentration and on the electrolyte nature. For 0.2 M KCl, the value of k, as follows from the results shown in Fig. 2, is equal to 1.123, while for an ideal solution with D = const, k = 1.06 according to Eq. (23).

By repeating the calculations for different KCl input concentrations (c_{0in} in mol dm^{-3}), we can obtain the $k(c)$ concentration dependence. Polynomial approximation of this dependence gives the following equation, valid in the concentration range from 0 to 0.5 M KCl:

$$\delta_{loc} = h\left(0.0206c_{0in}^2 + 0.0094c_{0in}^{3/2} - 0.1537c_{0in} + 0.184c_{0in}^{1/2} + 1.064\right)Y^{1/3} \tag{31}$$

Due to the fact that, generally, $g(c)$ depends only weakly on the concentration, δ_{loc} as well, slightly differs from the value related to the case D = const. Accordingly, the value of k in equation $\delta_{loc} = hkY^{1/3}$ is not very different from value 1.06, related to infinite dilution: in the case of KCl and c_{0in} = 0.2 mol/dm^3, k = 1.12; for c_{0in} = 0.5 mol dm^{-3}, k = 1.13.

Let us emphasize that the form of the Peers equation does not change, when we consider the concentration dependence $D(c)$ or not. It is due to the fact that this equation follows from Eq. (13), which is general.

Although the numerical variation of the limiting current value after the correction allowing for the concentration dependence of the diffusion coefficient is relatively low, the analysis above is important because it removes some difficulties in interpretation of the Peers equation and gives a better understanding of the role of the Nernst diffusion layer thickness.

4. Accounting for dependence D(c) in Sand equation

The Sand equation, Eq. (3), is deduced [15] when considering 1D diffusion in a semi-infinite region. A DC current density i is applied at $t = 0$, concentration and electric field variations in time caused by this current are the subjects of study. To describe these variations, the Fick's second law is used in the form:

$$\frac{\partial c}{\partial t} = \frac{\partial}{\partial x}\left(D(c)\frac{\partial c}{\partial x}\right), \tag{32}$$

obtained by substituting Eq. (15) into Eq. (7).

In the case of $D = \text{const}$, Eq. (32) has an analytical solution [12,13]. The use of boundary condition (13) at the membrane surface, the condition of constant concentration at a sufficiently great distance from the membrane surface

$$c(-\infty, t) = c_0, \tag{33}$$

and the initial condition

$$c(x, 0) = c_0, \tag{34}$$

lead to the following expression for the electrolyte concentration at the membrane surface as a function of time:

$$c(0,t) = c_0 - \frac{i}{z_1 F D}(T_1 - t_1)2\sqrt{\frac{Dt}{\pi}}. \tag{35}$$

As can be seen from Eq. (35), $c(0,t)$ decreases with increasing time. The transition time, τ, can be defined as the time $t = \tau$, when $c(0,t)$ vanishes. Setting $c(0,t) = 0$ in Eq. (35), we arrived at Eq. (3). Note that useful analytical expressions in chronopotentiometry are deduced by Sistat and Pourcelly [49] at underlimiting currents. A similar model is developed for chronoamperometry by Moya and Sistat [50].

At $t = \tau$, according to model (32)–(34), the potential drop in the system, $\Delta\varphi$, tends to infinity. In real systems, even at current densities essentially higher than i_{lim}, $c(0, \tau)$ takes small but nonzero values [51–53]. When $t \to \tau$, $\Delta\phi$ increases sharply, but does not tend to infinity. Instead, additional modes of ion transfer occur, which mitigate concentration polarization. Together with electrodiffusion of salt ions, the principle current transfer mode in the range of underlimiting currents, new charge carriers, H^+ and OH^- ions generated by water splitting, make their contribution [16,22]. Moreover, current-induced convection (gravitational convection and electroconvection) facilitates the delivery of fresh solution to the membrane surface [22,23,25,54,55], which reduces the resistance of the depleted solution at a given current density to a finite value, which determines the steady state of the system.

In the case of variable diffusion coefficient, the set of Eqs. (32), (17), (18) together with Eqs. (28)–(30) giving the $D(c)$ function, should be used. The boundary and initial conditions remain the same as in the case of $D = \text{const}$. However, in order to correctly determine the transition time, which occurs when the salt concentration at the surface is very small, but non-zero, the coupled Nernst-Planck-Poisson equations should be applied [51–53]. This problem has no analytical solution. However, it can be solved numerically using the finite difference method described in [48,56], see Supplementary Materials.

Numerical solution of the transition problem needs the use of a finite distance where the bulk concentration is set. Hence, instead of Eq. (33) we use the following condition:

$$c(-\delta_{\text{st}}, t) = c_0, \tag{36}$$

where δ_{st} is the DBL at steady state. It is the limit of current value of δ, when $t \to \infty$. δ_{st} determines the steady state potential difference in the system.

To find a numerical solution consistent with the Sand equation, the conditions required when obtaining the Sand equation should be satisfied, in particular,

$$\frac{\partial c(-\infty, t)}{\partial t} = 0, \tag{37}$$

This condition in our case of numerical calculation transforms in condition of vanishing the derivative $\partial c(-\delta_{\text{st}}, t)/\partial t = 0$ at $x = \delta_{\text{st}}$. For this, the current density must be essentially higher than i_{lim}. In this case, the transition time, τ, is sufficiently small, so that the current value $\delta(\tau)$ is considerably less than δ_{st}.

If the conditions above are satisfied, the transition time calculated numerically corresponds to the inflection point of the chronopotentiogram (Fig. 3a), which can be found by the maximum of derivative $d\Delta\varphi/dt$ (Fig. 3b). Since $(c_{1s}^{\text{I}})_{t\to\infty}$ is much less than c_{10}, the time needed to reach $(c_{1s}^{\text{I}})_{t\to\infty}$ by the boundary concentration is quite close to the time needed to reach zero boundary concentration, as it is required by the Sand model. Thus, the Sand transition time (τ_{Sand}) is very close to the transition time found by numerical computation (τ_{numcalc}), when the diffusion coefficient is assumed constant, $D = D_0$, and T_1 is taken equal to 1. Indeed, in the case of 0.2 M NaCl and an anion-exchange membrane, at $i = 47.7\ \text{mA cm}^{-2}$, $\tau_{\text{Sand}} = 13.24$ s and $\tau_{\text{numcalc}} = 13.25$ s. However, the number of nodes, N, dividing the depleted DBL in segments, should be rather high (1200). When $N = 600$, $\tau_{\text{numcalc}} = 13.30$ s. In this example, $\delta_{\text{st}} = 260\ \mu\text{m}$ that determines $i_{\text{lim}} = 30\ \text{mA cm}^{-2}$.

When the concentration dependence $D(c)$ is accounted for by using Eqs. (17), (18), all current values of D are lower than the value related to infinite dilution, D_0. This leads to smaller values of the transition time compared to the case $D = D_0 = \text{const}$ (Fig. 3), as predicts the Sand equation (3). It is of interest to find an effective value of the diffusion coefficient, D_{ef}, whose substitution into Eq. (3) gives the "correct" value of t, that is the value obtained by numerical calculation using concentration dependence $D(c)$. D_{ef} can be found from Eq. (3), as the value, which provides the same value of t as when using the numerical calculation with $D(c)$. Repeating this operation for different bulk concentrations, c_0, we find a series of values D_{ef}, as a function of c_0. In the case of NaCl, this function can be approximated by a polynomial:

$$D_{\text{ef}} = D_0\left(-0.0534c_0^{3/2} + 0.257c_0 - 0.260c_0^{1/2} + 1.00\right), \tag{38}$$

where $D_0 = 1.62\ 10^{-5}\ \text{cm}^2\text{s}^{-1}$ is the NaCl diffusion coefficient at infinite dilution.

Note that up to 0.01 M, the activity coefficients of any 1:1 strong electrolyte have the same concentration dependence that is explained by the Debye-Hückel theory. Accordingly, Eq. (38) will be valid for any 1:1 strong electrolyte in the range up to 0.01 M concentration.

The experiment was carried out using the cell and operating conditions described in [57]. We measured chronopotentiograms with NaCl solutions of three different concentrations: 0.002, 0.02 and $0.2\ \text{mol L}^{-1}$. The applied current density at each concentration was approximately 2 times higher than the limiting current density. Note that when the concentration and the current density increase both in 10 times, the transition time calculated according to the Sand equation, Eq. (3), remains constant under the condition that D and T_1 are fixed. In experiment, however, τ

Table 1
Concentration dependence of effective diffusion coefficient in the Sand equation (D_{ef}), counterion transport number in the membrane (T_1), and experimental (τ_{exp}) and calculated (τ_{calc}) transition times.

c_0, mol L^{-1}	D_{ef}, 10^{-5} cm^2 s^{-1}	T_1	i, mA cm^{-2}	τ_{exp}, s	τ_{calc}, s $D=D(c)$	τ_{calc}, s $D=D_0$
0.002	1.60	1	0.477	13.1	13.1	13.25
0.02	1.56	0.995	4.77	13.15	13.15	13.6
0.2	1.50	0.985	47.7	13.3	13.3	14.3

increases with increasing concentration (Table 1). This is due to the fact that with increasing concentration, the permselectivity of the membrane (characterized by T_1) decreases [2].

Fig. 3a shows experimental and theoretical chronopotentiograms in the case of 0.2 M NaCl. The $d\Delta\phi/dt$ derivatives are shown in Fig. 3b. The input (c_0, D_{ef}, T_1, i) and output (τ_{calc}) parameters together with experimental values of τ are presented in Table 1.

The theoretical curves were calculated with and without taking into account $D(c)$ dependence. The counterion transport numbers in the membrane were fitted in order to obtain in calculations with $D=D(c)$ the same value of τ as in the experiment. The values of T_1 found in this way (Table 1) show high permselectivity of the used AMX membrane that agrees with estimations reported for commercial homogeneous membranes by other publications [6,7,58]. It is seen that when using the $D=D_0$ approximation, the theoretical transition time is noticeably higher than the experimental one. This difference increases with increasing concentration. At $c_0=0.2$ mol L^{-1} $\tau_{exp}=13.3$ s, while $\tau_{calc}=14.3$ s. If we fit the values of T_1 to obtain in the simulations the experimental values of τ_c under the assumption of $D=D_0$, we find $T_1=1.002$, 1.001 and 0.999 for 0.002 M, 0.02 M and 0.2 M, respectively. We see that the $D=D_0$ assumption leads to unreasonable values of T_1. Even a small correction of the value of D results in a noticeable variation in T_1 (T_2), when the Sand equation is used for determination of ion transport numbers by using experimental transition time. Thus, taking into account the concentration dependence D(c) seems quite important when treating experimental chronopotentiograms [31,59,60,61].

5. Conclusion

In this paper, we investigate the possibility to find the limiting current density in voltammetry and the transition time in chronopotentiometry in the case, where the concentration dependence of diffusion coefficient is taken into account. We study two models with variable $D=D(c)$, which are more general than the fundamental models providing the Peers and the Sand equations, respectively. By applying a 2D model for calculating i_{lim}, we show that the numerical solution can be approximately presented in the form of the Peers equation. Similarly, a 1D non-stationary model with variable $D=D(c)$ allows obtaining an approximation for calculating τ in the form of Sand equations. Since with approaching the limiting current, the concentration at the depleted interface becomes very small, the diffusion coefficient in the Peers equation should be taken as at infinite dilution. At the same time, the Nernst diffusion layer thickness under given hydrodynamic conditions is a function of concentration. In the case of laminar flow between two smooth membranes without separator, the numerical solution gives the same shape of dependence of δ on the dimensionless channel length, Y, as in the case of $D=$const expressed using the well-known Lévêque equation: $\delta = kY^{1/3}$. However, the value of the proportionality factor k in the case of variable diffusion coefficient is a function of the input solution concentration. k (and δ) increase with increasing solution concentration, which is equivalent to a reduction in the mass transfer coefficient.

In the case of chronopotentiometry, we show that the results of numerical computation of the transition time can be presented in the form of the Sand equation, where an effective diffusion coefficient, D_{ef}, is introduced. D_{ef} takes some mean value (rather close to the arithmetic average) between D_0, related to the infinitely dilute solution, and D_b, related to the bulk solution. It is found that the account for the concentration dependence of the diffusion coefficient can significantly improve the agreement between the calculated and experimental chronopotentiograms in the range of concentrations >0.02 mol dm^{-3}.

Acknowledgments

The work is carried out at the French–Russian laboratory “Ion-Exchange Membranes and Related Processes”. We are grateful to the Russian Science Foundation, Russia (grant No. 14-19-00401, development of 1D model) and Russian Foundation for basic Research, Russia (grant No. 16-38-60135 mol_a_dk, chronopotentiometric measurements) for financial support, and to M.A. Vorotyntsev for helpful discussions.

Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at http://dx.doi.org/10.1016/j.electacta.2016.02.098.

References

[1] A.M. Peers, Disc Faraday Soc 21 (1956) 124–125.
[2] F. Hellferich, Ion exchange, Dover Publications, New York, 1962.
[3] J. Manzanares, K. Kontturi Encyclopedia of Electrochemistry, Interfacial Kinetics and Mass Transport, in: Bard AJ, M., Stratmann, E.J. Calvo (Eds.), VCH-Wiley, Weinheim, 2003.
[4] S.-T. Hwang, K. Kammermeyer, Membranes in Separation, Wiley, New York, 1975.
[5] V.V. Nikonenko, A.B. Yaroslavtsev, G. Pourcelly, Ion transfer in and through charged membranes, in: A. Ciferri, A. Perico (Eds.), Structure, Properties, Theory. Chapter 9 in Ionic interactions in natural and synthetic macromolecules, John Wiley & Sons Inc., 2012, pp. 267–336.
[6] R. Lteif, L. Dammak, C. Larchet, B. Auclair, Détermination du nombre de transport d'un contre-ion dans une membrane échangeuse d'ions en utilisant la méthode de la pile de concentration, Eur. Polym. J. 37 (2001) 627–639.
[7] C. Larchet, B. Auclair, V. Nikonenko, Approximate evaluation of water transport number in ion-exchange membranes, Electrochim. Acta 49 (2004) 1711–1717.
[8] N.W. Rosenberg, C.E. Tirrell, Limiting currents in membrane cells, Ind. Eng. Chem. 49 (4) (1957) 780–784.
[9] E. Volodina, N. Pismenskaya, V. Nikonenko, C. Larchet, G. Pourcelly, Ion transfer across ion-exchange membranes with homogeneous and heterogeneous surfaces, J. Colloid Interface Sci. 285 (2005) 247–258.
[10] H. Hamani, R. Bouamrane, M. Kameche, C. Innocent, Z. Derriche, Transport number and current–voltage of a cation exchange membrane equilibrated in aqueous and organic solutions, Phys. Chem. Liq. 51 (3) (2013) 265–280.
[11] J.-H. Choi, H.-J. Lee, S.-H. Moon, Effects of electrolytes on the transport phenomena in a cation-exchange membrane, J. Colloid Interface Sci. 238 (1) (2001) 188–195.
[12] D. Lerche, H. Wolf, Quantitative characterisation of current-induced diffusion layers at cation-exchange membranes. I. Investigations of temporal and local behaviour of concentration profile at constant current density, Bioelectrochem Bioenerg 2 (4) (1975) 293–302.
[13] J.J. Krol, M. Wessling, H. Strathmann, Chronopotentiometry and overlimiting ion transport through monopolar ion exchange membranes, J. Membr. Sci. 162 (1999) 155–164.
[14] R. Ibanez, D.F. Stamatialis, M. Wessling, Role of membrane surface in concentration polarization at cation exchange membranes, J. Membr. Sci. 239 (1) (2004) 119–128.
[15] H.J.S. Sand III, On the concentration at the electrodes in a solution, with special reference to the liberation of hydrogen by electrolysis of a mixture of copper sulphate and sulphuric acid, Phil Mag S 1 (1) (1901) 45–79.
[16] R. Simons, Electric field effects on proton transfer between ionizable groups and water in ion exchange membranes, Electrochim. Acta 29 (2) (1984) 151–158.

[17] J.J. Krol, M. Wessling, H. Strathmann, Concentration polarization with monopolar ion exchange membranes: current–voltage curves and water dissociation, J. Membr. Sci. 162 (1999) 145–154.
[18] Y. Tanaka, Acceleration of water dissociation generated in an ion exchange membrane, J. Membr. Sci. 303 (2007) 234–243.
[19] V.G. Levich, Physicochemical hydrodynamics, Prentice Hall, Englewood Cliffs, NJ, 1962.
[20] E.I. Belova, G.Yu. Lopatkova, N.D. Pismenskaya, V.V. Nikonenko, C. Larchet, G. Pourcelly, The effect of anion-exchange membrane surface properties on mechanisms of overlimiting mass transfer, J. Phys. Chem. B 110 (2006) 13458–13469.
[21] V.M. Volgin, A.D. Davydov, Calculation of limiting current density of metal electrodeposition on vertical plane electrode under conditions of natural convection, Electrochim. Acta 49 (3) (2004) 365–372.
[22] V.V. Nikonenko, N.D. Pismenskaya, E.I. Belova, Ph. Sistat, P. Huguet, G. Pourcelly, C. Larchet, Intensive current transfer in membrane systems: Modelling, mechanisms and application in electrodialysis, Advances in Colloid and Interface Science 160 (1) (2010) 101–123.
[23] I. Rubinstein, B. Zaltzman, Electro-osmotically induced convection at a permselective membrane, Phys. Rev. E 62 (2) (2000) 2238–2251.
[24] N.A. Mishchuk, Concentration polarization of interface and non-linear electrokinetic phenomena, Adv Colloid Interface Sci 160 (1) (2010) 16–39.
[25] R. Kwak, V.S. Pham, K.M. Lim, J. Han, Shear Flow of an Electrically Charged Fluid by Ion Concentration Polarization: Scaling Laws for Electroconvective, Physical review letters 110 (11) (2013) 114501 (1–5).
[26] M. Wessling, L. Garrigós Morcillo, S. Abdu, Nanometer-thick lateral polyelectrolyte micropatterns induce macrosopic electro-osmotic chaotic fluid instabilities, Scientific reports 4 (6) (2014) 4294 (1–5).
[27] M.S. Chao, The diffusion coefficients of hypochlorite hypochlorous acid, and chlorine in aqueous media by chronopotentiometry, J. Electrochem. Soc. 115 (1968) 1172–1174.
[28] M.R. Bermejo, J. Gomez, A.M. Martinez, E. Barrado, Y. Castrillejo, Electrochemistry of terbium in the eutectic LiCl–KCl, Electrochim. Acta 53 (16) (2008) 5106–5112.
[29] X.T. Le, P. Viel, D.P. Tran, F. Grisotto, S. Palacin, Subscriber access provided by CEA, J. Phys. Chem. B 113 (17) (2009) 5829–5836.
[30] K. Vetter, Electrochemical kinetics: theoretical and experimental aspects, Academic Press, 1967.
[31] P. Długołęcki, B. Anet, S.J. Metz, K. Nijmeijer, M. Wessling, Transport limitations in ion exchange membranes at low salt concentrations, J. Membr. Sci. 346 (2010) 163–171.
[32] P. Kubíček, Generalization of the Cottrell's equation by taking into account the concentration dependence of the coefficient of diffusion, J. Electroanal. Chem. 78 (1977) 161–165.
[33] P. Kubíček, Generalization of the Sand equation taking into account the concentration dependence of diffusion coefficient, Collect. Czech Chem. Commun. 42 (1977) 2821–2831.
[34] I. Rubinstein, B. Zaltzman, A. Futerman, V. Gitis, V. Nikonenko, Reexamination of electrodiffusion time scales, Phys. Rev. E 79 (2009) 021506.
[35] A.A. Moya, Theory of the formation of the electric double layer at the ion exchange membrane–solution interface, Phys. Chem. Chem. Phys. 17 (2015) 5207–5218.
[36] A.A. Moya, The differential capacitance of the electric double layer in the diffusion boundary layer of ion-exchange membrane systems, Electrochim. Acta 178 (2015) 249–258.
[37] J.S. Newman, Electrochemical systems, Prentice Hall, Englewood Cliffs, NJ, 1973.
[38] R.B. Bird, W.E. Stewart, Lightfoot EN Transport Phenomena, J. Wiley & Sons, New Jersey, 2007.
[39] V.V. Nikonenko, A.V. Kovalenko, M.K. Urtenov, N.D. Pismenskaya, J. Han, P. Sistat, G. Pourcelly, Desalination at overlimiting currents: State-of-the-art and perspectives, Desalination 342 (2014) 85–106.
[40] A.A. Sonin, R.F. Probstein, A hydrodynamic theory of desalination by electrodialysis, Desalination 5 (3) (1968) 293–329.
[41] N.P. Gnusin, V.I. Zabolotsky, V.V. Nikonenko, M.Kh. Urtenov, Convective-diffusion model of the electrodialytic desalination. Limiting current and diffusion layer, Sov. Electrochem. 22 (1986) 273–278 (Elektrokhimiya 22, (1986) 298).
[42] S.V. Pham, Z. Li, K.M. Lim, J.K. White, J. Han, Direct numerical simulation of electroconvective instability and hysteretic current-voltage response of a permselective membrane, Phys. Rev. E 86 (4) (2012) 046310.
[43] M.K. Urtenov, A.M. Uzdenova, A.V. Kovalenko, V.V. Nikonenko, N.D. Pismenskaya, V.I. Vasil'eva, P. Sistat, G. Pourcelly, Basic mathematical model of overlimiting transfer enhanced by electroconvection in flow-through electrodialysis membrane cells, J. Membr. Sci. 447 (2013) 190–202.
[44] R.A. Robinson, R.H. Stokes, Electrolyte solution, 2nd ed., Dover Publications, New York, 2002.
[45] V.A. Shaposhnik, V.A. Kuzminykh, O.V. Grigorchuk, V.I. Vasil'eva, Analytical model of laminar flow electrodialysis with ion-exchange membranes, J. Membr. Sci. 133 (1997) 27–37.
[46] M.A. Lévêque, Les lois de la transmission de chaleur par convection, Ann. Min. 13 (1928) 201–299.
[47] D.G. Miller, Application of irreversible thermodynamics to electrolyte solutions. I. Determination of Ionic Transport Coefficients lij for isothermal vector transport processes in binary electrolyte systems, J. Phys. Chem. 70 (1966) 2639–2659.
[48] C. Larchet, S. Nouri, B. Auclair, L. Dammak, V. Nikonenko, Application of chronopotentiometry to determine the thickness of diffusion layer adjacent to an ion-exchange membrane under natural convection, Adv. Colloid Interface Sci. 139 (1) (2008) 45–61.
[49] P. Sistat, G. Pourcelly, Chronopotentiometric response of an ion exchanges membrane in the underlimiting current range. Transport phenomena within the diffusion layers, J. Membr. Sci. 123 (1997) 121–131.
[50] A.A. Moya, P. Sistat, Chronoamperometric response of ion-exchange membrane systems, J Membr Sci 444 (2013) 412–419.
[51] I. Rubinstein, L. Shtilman, Voltage against current curves of cation exchange membranes, J. Chem Soc Faraday Trans 75 (1979) 231–246.
[52] K.T. Chu, M.Z. Bazant, Electrochemical thin films at and above the classical limiting current, SIAM J. Appl. Math. 65 (5) (2005) 1485–1505.
[53] M.A. Urtenov, E.V. Kirillova, N.M. Seidova, V.V. Nikonenko, Decoupling of the Nernst-Planck and Poisson equations. Application to a membrane system at overlimiting currents, J. of Physical Chemistry B 111 (51) (2007) 14208–14222.
[54] E.K. Zholkovskij, M.A. Vorotyntsev, E. Staude, Electrokinetic instability of solution in a plane-parallel electrochemical cell, J Colloid Interface Sci 181 (1) (1996) 28–33.
[55] I. Rubinstein, B. Zaltzman, Equilibrium Electroconvective Instability, Phys Rev Letters 114 (11) (2015) 114502 (1–5).
[56] J. Garrido, S. Mafé, J. Pellicer, Generalization of a finite-difference numerical method for the steady-state and transient solutions of the Nernst-Planck flux equations, J. Membr. Sci. 24 (1985) 7–14.
[57] N. Pismenskaia, Ph. Sistat, P. Huguet, V. Nikonenko, G. Pourcelly, Chronopotentiometry applied to the study of ion transfer through anion exchange membranes, J. Membr. Sci. 228 (2004) 65–76.
[58] Tuan Le Xuan, C. Buess-Herman, H.D. Hurwitz, Absorption equilibrium and permselectivity of cation exchange membranes in sulfuric acid, sodium chloride and nickel sulfate media, J. Membr. Sci. 323 (2008) 288–298.
[59] K. Kontturi, S. Mafe, J.A. Manzanares, G. Sundholm, R. Vapola, Counterion transport numbers of poly(acrylic acid)-grafted porous ion-exchange membranes as determined from current step measurements, Electrochim. Acta 42 (1997) 2569–2575.
[60] A. Yamauchi, Ion transport through diffusion layer controlled by charge mosaic membrane, Int. J. Chem. Eng. (2012), doi:http://dx.doi.org/10.1155/2012/417179.
[61] A.M. Bernardes, M.A.S. Rodrigues, J.Z. Ferreira, Electrodialysis and Water Reuse: Novel Approaches, Springer Science & Business Media, New York, 2013.